Multi-Corner Static Timing Analysis Report

Design: VSDBabySoC | Platform: SKY130 HD

Clock constraint: 45 ns | Parasitics: global-routing estimates

16-Corner Timing Summary

Corner	Setup Slack (ns)	Hold Slack (ns)	TNS (ns)	WNS (ns)
ff_100C_1v65	41.11	0.26	0.00	0.00
ff_100C_1v95	42.03	0.21	0.00	0.00
ff_n40C_1v56	39.72	0.31	0.00	0.00
ff_n40C_1v65	40.49	0.27	0.00	0.00
ff_n40C_1v76	41.16	0.24	0.00	0.00
ff_n40C_1v95	41.91	0.20	0.00	0.00
ss_100C_1v40	30.11	0.93	0.00	0.00
ss_100C_1v60	35.00	0.66	0.00	0.00
ss_n40C_1v28	1.22	1.83	0.00	0.00
ss_n40C_1v35	15.51	1.37	0.00	0.00
ss_n40C_1v40	21.46	1.14	0.00	0.00
ss_n40C_1v44	24.95	1.01	0.00	0.00
ss_n40C_1v60	32.75	0.68	0.00	0.00
ss_n40C_1v76	36.41	0.51	0.00	0.00
tt_025C_1v80	40.00	0.33	0.00	0.00
tt_100C_1v80	40.13	0.33	0.00	0.00

Note: Results are from the current design database using a 45 ns clock constraint and global-routing parasitic estimates. They are not guaranteed to match another run.

Detailed Timing Reports

Corner: ff_100C_1v65

```
CORNER: ff_100C_1v65
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23    0.28 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01    0.29 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.27    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.27    0.01    0.61 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.07    0.19    0.80 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.00    0.81 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.26    1.06 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.00    1.06 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.31    0.28    1.35 ^ place489/X (sky130_fd_sc_hd__buf_4)
                  0.31    0.01    1.35 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.16    0.26    1.61 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                  0.16    0.00    1.61 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.11    0.13    1.74 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.11    0.00    1.74 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.08    0.35    2.10 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                  0.08    0.00    2.10 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.55    0.45    2.54 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.55    0.00    2.54 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.12    0.11    2.66 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.12    0.00    2.66 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.05    0.16    2.82 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.05    0.00    2.82 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.19    0.18    3.00 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                  0.19    0.00    3.00 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.13    0.11    3.11 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.13    0.00    3.11 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.54    0.49    3.60 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  0.54    0.00    3.60 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.17    0.15    3.75 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.17    0.00    3.75 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.70    0.59    4.34 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  0.70    0.00    4.34 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.17    0.20    4.54 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.17    0.00    4.54 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  4.54   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23   45.28 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01   45.29 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25    0.23    0.29   45.58 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01   45.58 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.06    0.18   45.76 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
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```
                  0.06    0.00   45.76 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   45.76   clock reconvergence pessimism
                         -0.12   45.65   library setup time
                                 45.65   data required time
-------------------------------------------------------------------------------
                                 45.65   data required time
                                 -4.54   data arrival time
-------------------------------------------------------------------------------
                                 41.11   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23    0.28 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01    0.29 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.27    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.27    0.01    0.61 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.06    0.19    0.80 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.00    0.80 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.25    1.05 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.00    1.05 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.05   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23    0.28 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01    0.29 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.27    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.27    0.01    0.61 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.07    0.19    0.81 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.00    0.81 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    0.81   clock reconvergence pessimism
                         -0.02    0.79   library hold time
                                  0.79   data required time
-------------------------------------------------------------------------------
                                  0.79   data required time
                                 -1.05   data arrival time
-------------------------------------------------------------------------------
                                  0.26   slack (MET)


=== SUMMARY ===
worst slack max 41.11
worst slack min 0.26
tns max 0.00
wns max 0.00
Clock clk
   0.74 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.59 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.15 setup skew
max slew
```

```
Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    0.91    0.60 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.03    0.02    0.02 (MET)
```

Corner: ff_100C_1v95

```
CORNER: ff_100C_1v95
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.17    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.23    0.26    0.50 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01    0.52 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.06    0.15    0.66 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.00    0.66 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.21    0.87 v core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.03    0.00    0.87 v place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.14    0.18    1.05 v place489/X (sky130_fd_sc_hd__buf_4)
                0.14    0.01    1.06 v _10805_/A (sky130_fd_sc_hd__ha_1)
     4    0.02    0.10    0.26    1.32 ^ _10805_/SUM (sky130_fd_sc_hd__ha_1)
                0.10    0.00    1.32 ^ _07452_/A1 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.10    0.11    1.42 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.10    0.00    1.42 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.07    0.27    1.69 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.07    0.00    1.69 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.45    0.35    2.04 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                0.45    0.00    2.04 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.11    0.07    2.11 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.11    0.00    2.11 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.04    0.13    2.24 v place329/X (sky130_fd_sc_hd__buf_4)
                0.04    0.00    2.24 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.15    0.14    2.38 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.15    0.00    2.38 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.12    0.09    2.47 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.12    0.00    2.47 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.43    0.37    2.84 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                0.43    0.00    2.84 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.15    0.10    2.94 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.15    0.00    2.94 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.55    0.45    3.39 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                0.55    0.00    3.39 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.15    0.13    3.52 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.15    0.00    3.52 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                3.52   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
```

```
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.17    0.20   45.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.17    0.01   45.25 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.26    0.20    0.23   45.48 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01   45.49 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.06    0.14   45.63 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.00   45.63 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   45.63   clock reconvergence pessimism
                         -0.09   45.54   library setup time
                                 45.54   data required time
-------------------------------------------------------------------------------
                                 45.54   data required time
                                 -3.52   data arrival time
-------------------------------------------------------------------------------
                                 42.03   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.17    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.23    0.26    0.50 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01    0.51 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.06    0.14    0.66 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.00    0.66 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.20    0.86 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.00    0.86 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.86   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.17    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.23    0.26    0.50 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01    0.52 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.06    0.15    0.67 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.00    0.67 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    0.67   clock reconvergence pessimism
                         -0.02    0.65   library hold time
                                  0.65   data required time
-------------------------------------------------------------------------------
                                  0.65   data required time
                                 -0.86   data arrival time
-------------------------------------------------------------------------------
                                  0.21   slack (MET)


=== SUMMARY ===
worst slack max 42.03
worst slack min 0.21
tns max 0.00
wns max 0.00
```

```
Clock clk
   0.62 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.50 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.12 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                  1.50    0.74    0.75 (MET)

max capacitance

Pin                                   Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                              0.05    0.02    0.03 (MET)
```

Corner: ff_n40C_1v56

```
CORNER: ff_n40C_1v56
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.25    0.29    0.33 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.25    0.01    0.34 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29    0.35    0.41    0.75 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.35    0.01    0.76 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.07    0.24    1.00 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    1.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.06    0.31    1.31 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.06    0.00    1.31 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.41    0.37    1.67 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.41    0.01    1.68 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.20    0.34    2.02 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.20    0.00    2.02 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.09    0.14    2.16 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.09    0.00    2.16 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.09    0.47    2.63 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.09    0.00    2.63 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.68    0.59    3.22 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                0.68    0.00    3.22 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.11    0.15    3.37 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.11    0.00    3.37 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.02    0.05    0.19    3.56 v place329/X (sky130_fd_sc_hd__buf_4)
                0.05    0.00    3.57 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.24    0.26    3.83 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.24    0.00    3.83 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.12    0.12    3.95 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.12    0.00    3.95 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.76    0.72    4.68 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                0.76    0.00    4.68 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.18    0.19    4.87 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
```

```
                0.18    0.00    4.87 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    1.00    0.89    5.76 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                1.00    0.00    5.77 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.17    0.28    6.04 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.17    0.00    6.04 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                6.04   data arrival time

                       45.00   45.00   clock clk (rise edge)
                        0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.25    0.29   45.33 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.25    0.01   45.34 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.24    0.29    0.37   45.71 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.29    0.01   45.72 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.07    0.23   45.94 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00   45.94 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00   45.94   clock reconvergence pessimism
                       -0.18   45.76   library setup time
                               45.76   data required time
-----------------------------------------------------------------------------
                               45.76   data required time
                               -6.04   data arrival time
-----------------------------------------------------------------------------
                               39.72   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.25    0.29    0.33 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.25    0.01    0.34 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29    0.35    0.41    0.75 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.35    0.01    0.76 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.07    0.24    0.99 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    1.00 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.30    1.30 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.05    0.00    1.30 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                1.30   data arrival time

                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.25    0.29    0.33 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.25    0.01    0.34 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29    0.35    0.41    0.75 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.35    0.01    0.76 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.08    0.25    1.00 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.08    0.00    1.01 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    1.01   clock reconvergence pessimism
                       -0.02    0.99   library hold time
                                0.99   data required time
-----------------------------------------------------------------------------
                                0.99   data required time
                               -1.30   data arrival time
```

```
---------------------------------------------------------------------------
                                0.31   slack (MET)


=== SUMMARY ===
worst slack max 39.72
worst slack min 0.31
tns max 0.00
wns max 0.00
Clock clk
   1.01 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.82 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.19 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    1.22    0.29 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.03    0.02    0.01 (MET)
```

Corner: ff_n40C_1v65

```
CORNER: ff_n40C_1v65
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.23    0.26    0.30 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01    0.31 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.36    0.67 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.31    0.01    0.68 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.07    0.21    0.89 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.89 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.27    1.16 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.05    0.00    1.16 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.36    0.32    1.48 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.36    0.01    1.49 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.18    0.29    1.78 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.18    0.00    1.78 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.08    0.12    1.90 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.08    0.00    1.90 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.08    0.40    2.31 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.08    0.00    2.31 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.59    0.51    2.82 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                0.59    0.00    2.82 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.10    0.12    2.94 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.10    0.00    2.94 v place329/A (sky130_fd_sc_hd__buf_4)
```

```
     5    0.03    0.04    0.17    3.11 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.04    0.00    3.11 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.21    0.23    3.34 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                  0.21    0.00    3.34 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.11    0.11    3.44 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.11    0.00    3.44 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.66    0.62    4.06 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  0.66    0.00    4.06 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.17    0.16    4.22 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.17    0.00    4.22 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.86    0.76    4.98 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  0.86    0.00    4.98 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.16    0.22    5.20 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.16    0.00    5.20 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  5.20   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.23    0.26   45.30 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01   45.31 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.24    0.26    0.33   45.64 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.26    0.01   45.64 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.07    0.20   45.84 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.00   45.84 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   45.84   clock reconvergence pessimism
                         -0.16   45.68   library setup time
                                 45.68   data required time
-----------------------------------------------------------------------------
                                 45.68   data required time
                                 -5.20   data arrival time
-----------------------------------------------------------------------------
                                 40.49   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.23    0.26    0.30 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01    0.31 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.36    0.67 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.31    0.01    0.68 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.06    0.20    0.88 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.00    0.89 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.26    1.15 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.00    1.15 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.15   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.23    0.26    0.30 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.23    0.01    0.31 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.36    0.67 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.31    0.01    0.68 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.07    0.21    0.89 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.89 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    0.89   clock reconvergence pessimism
                       -0.02    0.88   library hold time
                                0.88   data required time
-----------------------------------------------------------------------------
                                0.88   data required time
                               -1.15   data arrival time
-----------------------------------------------------------------------------
                                0.27   slack (MET)


=== SUMMARY ===
worst slack max 40.49
worst slack min 0.27
tns max 0.00
wns max 0.00
Clock clk
   0.89 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.73 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.16 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    1.08    0.42 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.03    0.02    0.02 (MET)
```

Corner: ff_n40C_1v76

```
CORNER: ff_n40C_1v76
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23    0.27 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.20    0.01    0.28 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.28    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.28    0.01    0.61 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.06    0.18    0.79 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.00    0.79 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.24    1.03 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.04    0.00    1.03 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.33    0.29    1.31 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.33    0.01    1.32 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.16    0.25    1.57 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
```

```
                0.16    0.00    1.57 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01  0.08    0.11    1.68 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.08    0.00    1.68 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01  0.07    0.34    2.02 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.07    0.00    2.02 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02  0.52    0.44    2.46 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                0.52    0.00    2.46 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00  0.10    0.09    2.55 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.10    0.00    2.55 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03  0.04    0.15    2.70 v place329/X (sky130_fd_sc_hd__buf_4)
                0.04    0.00    2.70 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01  0.18    0.20    2.90 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.18    0.00    2.90 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01  0.10    0.09    2.99 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.10    0.00    2.99 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02  0.57    0.53    3.52 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                0.57    0.00    3.52 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01  0.15    0.12    3.64 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.15    0.00    3.64 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04  0.74    0.65    4.29 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                0.74    0.00    4.29 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01  0.14    0.17    4.46 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.14    0.00    4.46 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                4.46   data arrival time

                       45.00   45.00   clock clk (rise edge)
                        0.00   45.00   clock source latency
     2    0.32  0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21  0.20    0.23   45.27 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.20    0.01   45.28 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25  0.23    0.29   45.57 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01   45.57 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05  0.06    0.17   45.74 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.00   45.75 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00   45.75   clock reconvergence pessimism
                       -0.13   45.62   library setup time
                               45.62   data required time
-----------------------------------------------------------------------------
                               45.62   data required time
                               -4.46   data arrival time
-----------------------------------------------------------------------------
                               41.16   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32  0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21  0.20    0.23    0.27 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.20    0.01    0.28 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30  0.28    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.28    0.01    0.61 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04  0.06    0.18    0.79 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.00    0.79 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00  0.04    0.23    1.02 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.04    0.00    1.02 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
```

```
                                1.02   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.20    0.23    0.27 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.20    0.01    0.28 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.28    0.32    0.60 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.28    0.01    0.61 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.07    0.18    0.79 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.79 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    0.79   clock reconvergence pessimism
                       -0.02    0.78   library hold time
                                0.78   data required time
-----------------------------------------------------------------------------
                                0.78   data required time
                               -1.02   data arrival time
-----------------------------------------------------------------------------
                                0.24   slack (MET)


=== SUMMARY ===
worst slack max 41.16
worst slack min 0.24
tns max 0.00
wns max 0.00
Clock clk
   0.80 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -0.66 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.14 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    0.95    0.55 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.04    0.02    0.02 (MET)
```

Corner: ff_n40C_1v95

```
CORNER: ff_n40C_1v95
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.18    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
```

```
    22    0.31    0.24    0.26    0.51 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.24    0.01    0.53 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.05    0.14    0.67 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.00    0.67 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.20    0.87 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.00    0.87 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.14    0.28    0.24    1.11 ^ place489/X (sky130_fd_sc_hd__buf_4)
                  0.28    0.01    1.12 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.14    0.20    1.32 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                  0.14    0.00    1.32 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.07    0.09    1.41 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.07    0.00    1.41 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.06    0.28    1.69 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                  0.06    0.00    1.69 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.45    0.37    2.06 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.45    0.00    2.06 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.09    0.06    2.11 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.09    0.00    2.11 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.04    0.13    2.24 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.04    0.00    2.24 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.15    0.16    2.40 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                  0.15    0.00    2.40 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.09    0.07    2.48 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.09    0.00    2.48 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.47    0.42    2.90 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  0.47    0.00    2.90 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.13    0.08    2.99 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.13    0.00    2.99 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.61    0.52    3.51 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  0.61    0.00    3.51 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.13    0.11    3.62 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.13    0.00    3.62 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  3.62   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.17    0.20   45.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.18    0.01   45.25 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25    0.20    0.24   45.49 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.20    0.01   45.50 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.05    0.14   45.63 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.00   45.64 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   45.64   clock reconvergence pessimism
                         -0.10   45.53   library setup time
                                 45.53   data required time
-----------------------------------------------------------------------------
                                 45.53   data required time
                                 -3.62   data arrival time
-----------------------------------------------------------------------------
                                 41.91   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
```

```
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.18    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.24    0.26    0.51 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.24    0.01    0.52 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.05    0.14    0.67 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.05    0.00    0.67 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.19    0.86 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.03    0.00    0.86 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                0.86   data arrival time

                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.17    0.20    0.24 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.18    0.01    0.25 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.24    0.26    0.51 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.24    0.01    0.53 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.06    0.15    0.67 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.00    0.67 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    0.67   clock reconvergence pessimism
                       -0.02    0.66   library hold time
                                0.66   data required time
-----------------------------------------------------------------------------
                                0.66   data required time
                               -0.86   data arrival time
-----------------------------------------------------------------------------
                                0.20   slack (MET)


=== SUMMARY ===
worst slack max 41.91
worst slack min 0.20
tns max 0.00
wns max 0.00
Clock clk
   0.62 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.51 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.11 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                  1.51    0.81    0.70 (MET)

max capacitance

Pin                                   Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                              0.05    0.02    0.03 (MET)
```

Corner: ss_100C_1v40

```
CORNER: ss_100C_1v40
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.45    0.58    0.63 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.45    0.01    0.64 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.62    0.89    1.53 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.62    0.01    1.54 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.14    0.60    2.14 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.14    0.00    2.15 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.12    0.97    3.11 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.12    0.00    3.11 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.76    0.78    3.89 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.76    0.01    3.90 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.38    0.89    4.79 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.38    0.00    4.79 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.31    0.44    5.22 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.31    0.00    5.22 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.28    1.78    7.01 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.28    0.00    7.01 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    1.73    1.55    8.55 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                1.73    0.00    8.56 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.33    0.79    9.35 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.33    0.00    9.35 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.12    0.53    9.88 v place329/X (sky130_fd_sc_hd__buf_4)
                0.12    0.00    9.88 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.48    0.54   10.42 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.48    0.00   10.42 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.36    0.53   10.96 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.36    0.00   10.96 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    1.60    1.58   12.54 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                1.60    0.00   12.54 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.43    0.81   13.35 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.43    0.00   13.35 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    2.13    1.92   15.27 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                2.13    0.00   15.27 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.40    1.06   16.33 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.40    0.00   16.33 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                        0.00   16.33   data arrival time

                       45.00   45.00   clock clk (rise edge)
                        0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.45    0.58   45.63 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.45    0.01   45.64 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25    0.52    0.82   46.46 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.52    0.01   46.47 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04    0.14    0.56   47.03 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.14    0.00   47.03 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00   47.03   clock reconvergence pessimism
                       -0.58   46.45   library setup time
                               46.45   data required time
---------------------------------------------------------------------------------
                               46.45   data required time
                              -16.33   data arrival time
---------------------------------------------------------------------------------
                               30.11   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
```

```
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.45    0.58    0.63 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.45    0.01    0.64 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.62    0.89    1.53 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.62    0.01    1.54 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.13    0.60    2.14 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.13    0.00    2.14 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.07    0.84    2.98 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.07    0.00    2.98 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                2.98   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.45    0.58    0.63 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.45    0.01    0.64 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.62    0.89    1.53 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.62    0.01    1.54 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.15    0.62    2.16 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.15    0.00    2.16 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    2.16   clock reconvergence pessimism
                         -0.12    2.05   library hold time
                                  2.05   data required time
---------------------------------------------------------------------------
                                  2.05   data required time
                                 -2.98   data arrival time
---------------------------------------------------------------------------
                                  0.93   slack (MET)


=== SUMMARY ===
worst slack max 30.11
worst slack min 0.93
tns max 0.00
wns max 0.00
Clock clk
   1.95 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.48 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.47 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
_08069_/Y                              9.97    2.51    7.46 (MET)

max capacitance

Pin                                   Limit     Cap   Slack
------------------------------------------------------------
_07625_/Y                              0.06    0.00    0.05 (MET)
```

Corner: ss_100C_1v60

```
CORNER: ss_100C_1v60
```

```
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.44    0.48 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.34    0.01    0.49 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.47    0.64    1.13 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.47    0.01    1.14 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.10    0.42    1.57 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.00    1.57 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.09    0.66    2.23 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.09    0.00    2.23 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.56    0.57    2.79 ^ place489/X (sky130_fd_sc_hd__buf_4)
                  0.56    0.01    2.80 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.29    0.61    3.41 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                  0.29    0.00    3.41 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.23    0.30    3.71 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.23    0.00    3.71 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.20    1.14    4.86 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                  0.20    0.00    4.86 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    1.21    1.06    5.92 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                  1.21    0.00    5.92 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.25    0.47    6.39 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.25    0.00    6.39 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.09    0.38    6.77 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.09    0.00    6.77 v _08170_/A1 (sky130_fd_sc_hd__o21a_1)
     2    0.01    0.10    0.38    7.15 v _08170_/X (sky130_fd_sc_hd__o21a_1)
                  0.10    0.00    7.15 v _08173_/A3 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.81    0.73    7.89 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.81    0.00    7.89 ^ _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.39    0.61    8.50 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  0.39    0.00    8.50 v _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.83    0.83    9.33 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.83    0.00    9.33 ^ _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    1.09    1.20   10.53 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  1.09    0.00   10.53 v _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.43    0.73   11.27 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.43    0.00   11.27 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.00   11.27   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.44   45.48 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.34    0.01   45.49 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25    0.39    0.59   46.08 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.39    0.01   46.09 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.10    0.40   46.48 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.00   46.49 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   46.49   clock reconvergence pessimism
                         -0.22   46.27   library setup time
                                 46.27   data required time
---------------------------------------------------------------------------
                                 46.27   data required time
                                -11.27   data arrival time
```

```
---------------------------------------------------------------------------
                              35.00   slack (MET)


=== HOLD ===
Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                 0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.44    0.48 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.34    0.01    0.49 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.47    0.64    1.13 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.47    0.01    1.14 ^ clkbuf_leaf_2_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    12    0.05    0.11    0.43    1.57 ^ clkbuf_leaf_2_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.11    0.00    1.57 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.59    2.16 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                 0.05    0.00    2.16 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
                                 2.16   data arrival time

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                 0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.44    0.48 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.34    0.01    0.49 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.47    0.64    1.13 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.47    0.01    1.14 ^ clkbuf_leaf_2_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    12    0.05    0.11    0.43    1.57 ^ clkbuf_leaf_2_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.11    0.00    1.57 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
                         0.00    1.57   clock reconvergence pessimism
                        -0.07    1.50   library hold time
                                 1.50   data required time
---------------------------------------------------------------------------
                                 1.50   data required time
                                -2.16   data arrival time
---------------------------------------------------------------------------
                                 0.66   slack (MET)


=== SUMMARY ===
worst slack max 35.00
worst slack min 0.66
tns max 0.00
wns max 0.00
Clock clk
   1.43 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.10 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.33 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
_08069_/Y                              1.47    1.70   -0.23 (VIOLATED)

max capacitance
```

```
Pin                                  Limit     Cap   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 0.10    0.12   -0.01 (VIOLATED)
```

Corner: ss_n40C_1v28

```
CORNER: ss_n40C_1v28
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    1.42    1.38    1.42 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.42    0.01    1.43 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.27    1.85    2.50    3.93 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.85    0.01    3.94 ^ clkbuf_leaf_54_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.37    1.66    5.60 ^ clkbuf_leaf_54_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.37    0.00    5.60 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.27    2.39    7.99 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.27    0.00    7.99 ^ place486/A (sky130_fd_sc_hd__buf_4)
    22    0.09    1.71    1.65    9.64 ^ place486/X (sky130_fd_sc_hd__buf_4)
                  1.71    0.00    9.64 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
     1    0.00    0.69    2.05   11.69 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
                  0.69    0.00   11.69 ^ place430/A (sky130_fd_sc_hd__buf_4)
    35    0.23    4.27    3.76   15.46 ^ place430/X (sky130_fd_sc_hd__buf_4)
                  4.27    0.02   15.47 ^ _07595_/S0 (sky130_fd_sc_hd__mux4_2)
     2    0.01    0.54    7.14   22.61 v _07595_/X (sky130_fd_sc_hd__mux4_2)
                  0.54    0.00   22.61 v _07789_/A1 (sky130_fd_sc_hd__mux2i_1)
     2    0.01    1.91    2.04   24.66 ^ _07789_/Y (sky130_fd_sc_hd__mux2i_1)
                  1.91    0.00   24.66 ^ _08184_/B1 (sky130_fd_sc_hd__o22ai_1)
     2    0.01    0.46    1.38   26.04 v _08184_/Y (sky130_fd_sc_hd__o22ai_1)
                  0.46    0.00   26.04 v _08185_/A2 (sky130_fd_sc_hd__a21oi_1)
     1    0.01    1.61    1.78   27.82 ^ _08185_/Y (sky130_fd_sc_hd__a21oi_1)
                  1.61    0.00   27.82 ^ _08186_/B (sky130_fd_sc_hd__nor2_1)
     1    0.00    0.41    0.89   28.71 v _08186_/Y (sky130_fd_sc_hd__nor2_1)
                  0.41    0.00   28.71 v _08187_/C1 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    5.37    4.68   33.38 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  5.37    0.00   33.38 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.86    2.81   36.19 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.86    0.00   36.19 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    7.59    6.80   42.99 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  7.59    0.00   42.99 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.00    0.79    4.10   47.09 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.79    0.00   47.09 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                 47.09   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    1.42    1.38   46.42 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.42    0.01   46.43 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.22    1.54    2.27   48.70 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.54    0.01   48.71 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04    0.37    1.48   50.19 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.37    0.00   50.19 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
                          0.00   50.19   clock reconvergence pessimism
                         -1.87   48.32   library setup time
                                 48.32   data required time
-------------------------------------------------------------------------------
                                 48.32   data required time
                                -47.09   data arrival time
-------------------------------------------------------------------------------
                                  1.22   slack (MET)


=== HOLD ===
Startpoint: core.CPU_src2_value_a3[12]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a4[12]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    1.42    1.38    1.42 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.42    0.01    1.43 ^ clkbuf_3_6__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.13    0.93    1.82    3.25 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.93    0.00    3.26 ^ clkbuf_leaf_57_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    12    0.04    0.35    1.12    4.37 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.35    0.00    4.38 ^ core.CPU_src2_value_a3[12]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     3    0.01    0.16    1.77    6.15 v core.CPU_src2_value_a3[12]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.16    0.00    6.15 v core.CPU_src2_value_a4[12]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  6.15   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    1.42    1.38    1.42 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.42    0.01    1.43 ^ clkbuf_3_7__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    10    0.14    1.02    1.89    3.32 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.02    0.00    3.33 ^ clkbuf_leaf_58_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    10    0.05    0.43    1.23    4.55 ^ clkbuf_leaf_58_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.43    0.00    4.56 ^ core.CPU_src2_value_a4[12]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    4.56   clock reconvergence pessimism
                         -0.24    4.31   library hold time
                                  4.31   data required time
-------------------------------------------------------------------------------
                                  4.31   data required time
                                 -6.15   data arrival time
-------------------------------------------------------------------------------
                                  1.83   slack (MET)


=== SUMMARY ===
worst slack max 1.22
worst slack min 1.83
tns max 0.00
wns max 0.00
Clock clk
   5.64 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -4.37 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   1.27 setup skew

max slew
```

```
Pin                                  Limit    Slew   Slack
------------------------------------------------------------
_08319_/Y                             4.97    7.66   -2.69 (VIOLATED)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 0.09    0.12   -0.02 (VIOLATED)
```

Corner: ss_n40C_1v35

```
CORNER: ss_n40C_1v35
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    0.97    1.00    1.04 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.97    0.01    1.06 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.29    1.76    2.81 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                1.29    0.01    2.82 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.25    1.14    3.96 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.25    0.00    3.96 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.20    1.65    5.61 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.20    0.00    5.61 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.12    1.56    1.45    7.06 ^ place489/X (sky130_fd_sc_hd__buf_4)
                1.56    0.01    7.07 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.67    1.59    8.66 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.67    0.00    8.66 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.28    0.60    9.26 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.28    0.00    9.26 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.43    3.24   12.50 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.43    0.00   12.50 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    3.14    2.89   15.39 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                3.14    0.00   15.39 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.36    1.52   16.91 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.36    0.00   16.91 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.02    0.13    0.83   17.74 v place329/X (sky130_fd_sc_hd__buf_4)
                0.13    0.00   17.75 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    1.05    1.18   18.92 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                1.05    0.00   18.92 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.33    0.82   19.75 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.33    0.00   19.75 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    3.60    3.42   23.17 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                3.60    0.00   23.17 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.62    1.78   24.95 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.62    0.00   24.95 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    5.01    4.52   29.48 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                5.01    0.00   29.48 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.00    0.56    2.46   31.94 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.56    0.00   31.94 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                31.94   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
```

```
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    0.97    1.00   46.04 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.97    0.01   46.06 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.23    1.07    1.60   47.65 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.07    0.01   47.66 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04    0.26    1.04   48.70 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.26    0.00   48.70 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   48.70   clock reconvergence pessimism
                         -1.25   47.45   library setup time
                                 47.45   data required time
-----------------------------------------------------------------------------
                                 47.45   data required time
                                -31.94   data arrival time
-----------------------------------------------------------------------------
                                 15.51   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    0.97    1.00    1.04 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.97    0.01    1.06 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.29    1.76    2.81 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.29    0.01    2.82 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.24    1.13    3.96 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.24    0.00    3.96 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.07    1.24    5.20 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.07    0.00    5.20 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  5.20   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.20    0.97    1.00    1.04 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.97    0.01    1.06 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.29    1.76    2.81 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  1.29    0.01    2.82 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.29    1.17    3.99 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.29    0.00    3.99 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    3.99   clock reconvergence pessimism
                         -0.16    3.83   library hold time
                                  3.83   data required time
-----------------------------------------------------------------------------
                                  3.83   data required time
                                 -5.20   data arrival time
-----------------------------------------------------------------------------
                                  1.37   slack (MET)


=== SUMMARY ===
worst slack max 15.51
worst slack min 1.37
tns max 0.00
wns max 0.00
```

```
Clock clk
   4.00 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -3.14 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.86 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
_08319_/Y                              9.97    5.22    4.75 (MET)

max capacitance

Pin                                   Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                              0.04    0.02    0.02 (MET)
```

Corner: ss_n40C_1v40

```
CORNER: ss_n40C_1v40
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                 0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.78    0.84    0.88 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.78    0.01    0.89 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.05    1.43    2.32 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 1.05    0.01    2.33 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.21    0.94    3.26 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.21    0.00    3.26 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.17    1.33    4.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                 0.17    0.00    4.60 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.12    1.28    1.20    5.79 ^ place489/X (sky130_fd_sc_hd__buf_4)
                 1.28    0.01    5.80 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.57    1.31    7.11 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                 0.57    0.00    7.11 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.25    0.51    7.61 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                 0.25    0.00    7.61 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.34    2.57   10.18 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                 0.34    0.00   10.18 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    2.55    2.34   12.53 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                 2.55    0.00   12.53 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.31    1.17   13.70 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                 0.31    0.00   13.70 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.02    0.12    0.69   14.39 v place329/X (sky130_fd_sc_hd__buf_4)
                 0.12    0.00   14.39 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.84    0.95   15.34 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                 0.84    0.00   15.34 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.30    0.67   16.02 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                 0.30    0.00   16.02 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    2.87    2.76   18.77 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                 2.87    0.00   18.77 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.50    1.38   20.15 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
```

```
                0.50    0.00   20.15 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    3.95    3.56   23.72 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                3.95    0.00   23.72 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.00    0.47    1.88   25.60 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.47    0.00   25.60 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                               25.60   data arrival time

                       45.00   45.00   clock clk (rise edge)
                        0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.78    0.84   45.88 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.78    0.01   45.89 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.23    0.88    1.30   47.19 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.88    0.01   47.19 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04    0.21    0.85   48.04 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.21    0.00   48.04 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00   48.04   clock reconvergence pessimism
                       -0.98   47.07   library setup time
                               47.07   data required time
-----------------------------------------------------------------------------
                               47.07   data required time
                              -25.60   data arrival time
-----------------------------------------------------------------------------
                               21.46   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.78    0.84    0.88 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.78    0.01    0.89 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.05    1.43    2.32 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                1.05    0.01    2.33 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.20    0.93    3.26 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.20    0.00    3.26 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.06    1.04    4.30 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.06    0.00    4.30 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                4.30   data arrival time

                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.78    0.84    0.88 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.78    0.01    0.89 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    1.05    1.43    2.32 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                1.05    0.01    2.33 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.24    0.96    3.29 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.24    0.00    3.29 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    3.29   clock reconvergence pessimism
                       -0.14    3.15   library hold time
                                3.15   data required time
-----------------------------------------------------------------------------
                                3.15   data required time
                               -4.30   data arrival time
```

```
-------------------------------------------------------------------------------
                                1.14   slack (MET)


=== SUMMARY ===
worst slack max 21.46
worst slack min 1.14
tns max 0.00
wns max 0.00
Clock clk
   3.29 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -2.58 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.71 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
_08319_/Y                             9.99    4.19    5.80 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.05    0.02    0.03 (MET)
```

Corner: ss_n40C_1v44

```
CORNER: ss_n40C_1v44
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.68    0.74    0.78 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.68    0.01    0.79 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    0.91    1.24    2.03 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.91    0.01    2.04 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.18    0.79    2.83 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.18    0.00    2.83 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.15    1.14    3.98 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.15    0.00    3.98 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.12    1.11    1.05    5.02 ^ place489/X (sky130_fd_sc_hd__buf_4)
                1.11    0.01    5.03 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.50    1.13    6.16 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.50    0.00    6.16 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.23    0.45    6.61 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.23    0.00    6.61 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.29    2.18    8.79 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.29    0.00    8.79 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    2.19    2.01   10.80 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                2.19    0.00   10.80 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.28    0.97   11.77 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.28    0.00   11.77 v place329/A (sky130_fd_sc_hd__buf_4)
```

```
     5    0.02    0.10    0.61   12.38 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.10    0.00   12.38 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.74    0.84   13.22 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                  0.74    0.00   13.22 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.27    0.58   13.80 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.27    0.00   13.80 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    2.46    2.36   16.17 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  2.46    0.00   16.17 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.44    1.15   17.32 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.44    0.00   17.32 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    3.34    3.02   20.34 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  3.34    0.00   20.34 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.00    0.40    1.55   21.89 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.40    0.00   21.89 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                 21.89   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.68    0.74   45.78 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.68    0.01   45.79 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.23    0.76    1.13   46.92 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.76    0.01   46.92 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04    0.18    0.73   47.65 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.18    0.00   47.65 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   47.65   clock reconvergence pessimism
                         -0.81   46.84   library setup time
                                 46.84   data required time
-----------------------------------------------------------------------------
                                 46.84   data required time
                                -21.89   data arrival time
-----------------------------------------------------------------------------
                                 24.95   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.68    0.74    0.78 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.68    0.01    0.79 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    0.91    1.24    2.03 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.91    0.01    2.04 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.17    0.79    2.83 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.17    0.00    2.83 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.06    0.91    3.74 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.06    0.00    3.74 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  3.74   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                  0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.68    0.74    0.78 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.68    0.01    0.79 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.28    0.91    1.24    2.03 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.91    0.01    2.04 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.21    0.82    2.85 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.21    0.00    2.85 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    2.85   clock reconvergence pessimism
                       -0.12    2.73   library hold time
                                2.73   data required time
-----------------------------------------------------------------------------
                                2.73   data required time
                               -3.74   data arrival time
-----------------------------------------------------------------------------
                                1.01   slack (MET)


=== SUMMARY ===
worst slack max 24.95
worst slack min 1.01
tns max 0.00
wns max 0.00
Clock clk
   2.86 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -2.26 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.59 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
_08319_/Y                             4.96    3.58    1.38 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.03    0.02    0.01 (MET)
```

Corner: ss_n40C_1v60

```
CORNER: ss_n40C_1v60
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.44    0.51    0.55 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.44    0.01    0.56 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29    0.61    0.81    1.37 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.61    0.01    1.38 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.12    0.52    1.90 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.12    0.00    1.90 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.10    0.72    2.62 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.10    0.00    2.62 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.74    0.70    3.32 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.74    0.01    3.32 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.35    0.73    4.06 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
```

```
                0.35    0.00    4.06 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01  0.17    0.30    4.36 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.17    0.00    4.36 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01  0.18    1.30    5.65 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.18    0.00    5.65 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02  1.40    1.26    6.91 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                1.40    0.00    6.91 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00  0.20    0.55    7.46 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.20    0.00    7.46 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.02  0.08    0.41    7.87 v place329/X (sky130_fd_sc_hd__buf_4)
                0.08    0.00    7.87 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01  0.44    0.52    8.39 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.44    0.00    8.39 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01  0.20    0.36    8.75 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.20    0.00    8.75 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02  1.53    1.47   10.22 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                1.53    0.00   10.22 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01  0.32    0.65   10.87 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.32    0.00   10.87 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04  2.04    1.84   12.72 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                2.04    0.00   12.72 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.00  0.29    0.85   13.57 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.29    0.00   13.57 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                        13.57   data arrival time

                       45.00   45.00   clock clk (rise edge)
                        0.00   45.00   clock source latency
     2    0.32  0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21  0.44    0.51   45.55 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.44    0.01   45.56 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.24  0.51    0.73   46.30 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.51    0.01   46.30 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.04  0.12    0.48   46.78 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.12    0.00   46.78 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00   46.78   clock reconvergence pessimism
                       -0.46   46.32   library setup time
                               46.32   data required time
-----------------------------------------------------------------------------
                               46.32   data required time
                              -13.57   data arrival time
-----------------------------------------------------------------------------
                               32.75   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32  0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21  0.44    0.51    0.55 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.44    0.01    0.56 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29  0.61    0.81    1.37 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.61    0.01    1.38 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04  0.12    0.52    1.90 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.12    0.00    1.90 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00  0.04    0.61    2.51 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.04    0.00    2.51 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
```

```
                                2.51   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.44    0.51    0.55 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.44    0.01    0.56 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.29    0.61    0.81    1.37 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.61    0.01    1.38 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.14    0.53    1.91 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.14    0.00    1.91 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    1.91   clock reconvergence pessimism
                       -0.09    1.83   library hold time
                                1.83   data required time
-----------------------------------------------------------------------------
                                1.83   data required time
                               -2.51   data arrival time
-----------------------------------------------------------------------------
                                0.68   slack (MET)


=== SUMMARY ===
worst slack max 32.75
worst slack min 0.68
tns max 0.00
wns max 0.00
Clock clk
   1.92 source latency core.CPU_src1_value_a3[9]$_DFF_P_/CLK ^
  -1.53 target latency core.CPU_src2_value_a3[18]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.39 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
_08319_/Y                             1.48    2.23   -0.75 (VIOLATED)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 0.08    0.12   -0.04 (VIOLATED)
```

Corner: ss_n40C_1v76

```
CORNER: ss_n40C_1v76
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.40    0.44 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.34    0.01    0.45 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
```

```
    22    0.30    0.46    0.60    1.05 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.46    0.01    1.06 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.04    0.09    0.37    1.43 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.09    0.00    1.43 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.08    0.52    1.95 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.08    0.00    1.95 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.55    0.52    2.47 ^ place489/X (sky130_fd_sc_hd__buf_4)
                  0.55    0.01    2.48 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.28    0.53    3.01 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                  0.28    0.00    3.01 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.14    0.22    3.24 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.14    0.00    3.24 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.13    0.89    4.13 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                  0.13    0.00    4.13 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    1.01    0.90    5.03 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                  1.01    0.00    5.03 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.16    0.35    5.38 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.16    0.00    5.38 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.02    0.06    0.31    5.69 v place329/X (sky130_fd_sc_hd__buf_4)
                  0.06    0.00    5.69 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.33    0.37    6.06 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                  0.33    0.00    6.06 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.17    0.27    6.33 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                  0.17    0.00    6.33 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    1.09    1.06    7.38 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                  1.09    0.00    7.38 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.24    0.41    7.80 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                  0.24    0.00    7.80 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    1.46    1.29    9.09 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                  1.46    0.00    9.09 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.23    0.54    9.64 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                  0.23    0.00    9.64 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  9.64   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                  0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.40   45.44 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.34    0.01   45.45 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.24    0.39    0.55   45.99 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.39    0.01   46.00 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.10    0.35   46.35 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.00   46.35 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   46.35   clock reconvergence pessimism
                         -0.30   46.05   library setup time
                                 46.05   data required time
-----------------------------------------------------------------------------
                                 46.05   data required time
                                 -9.64   data arrival time
-----------------------------------------------------------------------------
                                 36.41   slack (MET)


=== HOLD ===
Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
```

```
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.40    0.44 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.34    0.01    0.45 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.46    0.60    1.05 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.46    0.01    1.05 ^ clkbuf_leaf_2_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    12    0.05    0.10    0.37    1.43 ^ clkbuf_leaf_2_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.10    0.00    1.43 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.46    1.89 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.03    0.00    1.89 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
                                1.89   data arrival time

                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.34    0.40    0.44 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.34    0.01    0.45 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.46    0.60    1.05 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.46    0.01    1.05 ^ clkbuf_leaf_2_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    12    0.05    0.10    0.37    1.43 ^ clkbuf_leaf_2_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.10    0.00    1.43 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
                        0.00    1.43   clock reconvergence pessimism
                       -0.05    1.38   library hold time
                                1.38   data required time
-----------------------------------------------------------------------------
                                1.38   data required time
                               -1.89   data arrival time
-----------------------------------------------------------------------------
                                0.51   slack (MET)


=== SUMMARY ===
worst slack max 36.41
worst slack min 0.51
tns max 0.00
wns max 0.00
Clock clk
   1.30 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -1.01 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.29 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    1.69   -0.18 (VIOLATED)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 0.10    0.12   -0.01 (VIOLATED)
```

Corner: tt_025C_1v80

```
CORNER: tt_025C_1v80
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.38    0.70 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.31    0.01    0.72 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.07    0.23    0.94 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.95 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.32    1.26 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.05    0.00    1.26 ^ place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.36    0.33    1.60 ^ place489/X (sky130_fd_sc_hd__buf_4)
                0.36    0.01    1.61 ^ _10805_/A (sky130_fd_sc_hd__ha_1)
     5    0.02    0.19    0.32    1.92 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
                0.19    0.00    1.92 ^ _07452_/A2 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.11    0.15    2.07 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                0.11    0.00    2.07 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.09    0.49    2.56 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                0.09    0.00    2.56 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.65    0.56    3.11 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                0.65    0.00    3.11 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.14    0.17    3.28 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                0.14    0.00    3.28 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.05    0.20    3.48 v place329/X (sky130_fd_sc_hd__buf_4)
                0.05    0.00    3.48 v _08172_/A1 (sky130_fd_sc_hd__o2111ai_1)
     1    0.01    0.22    0.23    3.71 ^ _08172_/Y (sky130_fd_sc_hd__o2111ai_1)
                0.22    0.00    3.71 ^ _08173_/B1 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.13    0.17    3.89 v _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                0.13    0.00    3.89 v _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.66    0.62    4.51 ^ _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                0.66    0.00    4.51 ^ _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.20    0.21    4.72 v _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                0.20    0.00    4.72 v _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.88    0.76    5.48 ^ _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                0.88    0.00    5.48 ^ _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.18    0.25    5.73 v _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                0.18    0.00    5.73 v core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                  5.73   data arrival time

                         45.00   45.00   clock clk (rise edge)
                          0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.22    0.27   45.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01   45.32 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.25    0.26    0.35   45.67 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.26    0.01   45.68 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.07    0.22   45.90 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00   45.90 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00   45.90   clock reconvergence pessimism
                         -0.16   45.73   library setup time
                                 45.73   data required time
---------------------------------------------------------------------------------
                                 45.73   data required time
                                 -5.73   data arrival time
---------------------------------------------------------------------------------
                                 40.00   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
```

```
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.38    0.70 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.31    0.01    0.71 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.04    0.07    0.23    0.94 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.94 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.31    1.25 ^ core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.04    0.00    1.25 ^ core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                1.25   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.21    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.23    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.30    0.31    0.38    0.70 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.31    0.01    0.71 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.08    0.24    0.95 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.08    0.00    0.95 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        0.00    0.95   clock reconvergence pessimism
                       -0.03    0.93   library hold time
                                0.93   data required time
---------------------------------------------------------------------------
                                0.93   data required time
                               -1.25   data arrival time
---------------------------------------------------------------------------
                                0.33   slack (MET)


=== SUMMARY ===
worst slack max 40.00
worst slack min 0.33
tns max 0.00
wns max 0.00
Clock clk
   0.87 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.69 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.18 setup skew

max slew

Pin                                  Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                 1.50    1.07    0.43 (MET)

max capacitance

Pin                                  Limit     Cap   Slack
------------------------------------------------------------
_08069_/Y                             0.03    0.02    0.01 (MET)
```

Corner: tt_100C_1v80

```
CORNER: tt_100C_1v80
```

```
PARASITICS: global_routing
=== SETUP ===
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                 0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.22    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.30    0.37    0.68 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.30    0.01    0.70 ^ clkbuf_leaf_49_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    14    0.05    0.07    0.22    0.92 ^ clkbuf_leaf_49_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.00    0.92 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.31    1.23 v core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                 0.04    0.00    1.23 v place489/A (sky130_fd_sc_hd__buf_4)
    24    0.13    0.18    0.25    1.48 v place489/X (sky130_fd_sc_hd__buf_4)
                 0.18    0.01    1.49 v _10805_/A (sky130_fd_sc_hd__ha_1)
     4    0.02    0.14    0.39    1.88 ^ _10805_/SUM (sky130_fd_sc_hd__ha_1)
                 0.14    0.00    1.88 ^ _07452_/A1 (sky130_fd_sc_hd__o21ai_0)
     2    0.01    0.13    0.16    2.04 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
                 0.13    0.00    2.04 v _07525_/B (sky130_fd_sc_hd__or3b_2)
     2    0.01    0.10    0.47    2.52 v _07525_/X (sky130_fd_sc_hd__or3b_2)
                 0.10    0.00    2.52 v _07531_/A1 (sky130_fd_sc_hd__o21ai_0)
     4    0.02    0.65    0.54    3.06 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
                 0.65    0.00    3.06 ^ _07955_/B2 (sky130_fd_sc_hd__a221oi_1)
     1    0.00    0.15    0.17    3.23 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
                 0.15    0.00    3.23 v place329/A (sky130_fd_sc_hd__buf_4)
     5    0.03    0.05    0.20    3.43 v place329/X (sky130_fd_sc_hd__buf_4)
                 0.05    0.00    3.43 v _08170_/A1 (sky130_fd_sc_hd__o21a_1)
     2    0.01    0.06    0.20    3.63 v _08170_/X (sky130_fd_sc_hd__o21a_1)
                 0.06    0.00    3.63 v _08173_/A3 (sky130_fd_sc_hd__o31ai_1)
     1    0.01    0.45    0.38    4.01 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
                 0.45    0.00    4.01 ^ _08187_/A2 (sky130_fd_sc_hd__a211oi_1)
     1    0.02    0.22    0.28    4.28 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
                 0.22    0.00    4.28 v _08188_/C1 (sky130_fd_sc_hd__a221oi_1)
     2    0.01    0.49    0.44    4.72 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
                 0.49    0.00    4.72 ^ _08192_/B1 (sky130_fd_sc_hd__o221ai_1)
    11    0.04    0.58    0.58    5.30 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
                 0.58    0.00    5.31 v _09262_/A1 (sky130_fd_sc_hd__o21ai_0)
     1    0.01    0.25    0.34    5.65 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
                 0.25    0.00    5.65 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                                 5.65   data arrival time

                        45.00   45.00   clock clk (rise edge)
                         0.00   45.00   clock source latency
     2    0.32    0.00    0.00   45.00 ^ pll/CLK (avsdpll)
                 0.08    0.04   45.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.22    0.27   45.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.22    0.01   45.32 ^ clkbuf_3_1__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    16    0.26    0.25    0.34   45.65 ^ clkbuf_3_1__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.25    0.01   45.66 ^ clkbuf_leaf_30_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    13    0.05    0.07    0.21   45.87 ^ clkbuf_leaf_30_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.00   45.87 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         0.00   45.87   clock reconvergence pessimism
                        -0.10   45.78   library setup time
                                45.78   data required time
---------------------------------------------------------------------------
                                45.78   data required time
                                -5.65   data arrival time
```

```
---------------------------------------------------------------------------
                                40.13   slack (MET)


=== HOLD ===
Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.22    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.30    0.37    0.68 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.30    0.01    0.69 ^ clkbuf_leaf_48_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.05    0.07    0.22    0.92 ^ clkbuf_leaf_48_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.00    0.92 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.31    1.22 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
                0.03    0.00    1.22 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                                1.22   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
     2    0.32    0.00    0.00    0.00 ^ pll/CLK (avsdpll)
                0.08    0.04    0.04 ^ clkbuf_0_CLK/A (sky130_fd_sc_hd__clkbuf_16)
     8    0.22    0.22    0.27    0.31 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.22    0.01    0.32 ^ clkbuf_3_2__f_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    22    0.31    0.30    0.37    0.68 ^ clkbuf_3_2__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.30    0.01    0.69 ^ clkbuf_leaf_47_CLK/A (sky130_fd_sc_hd__clkbuf_16)
    11    0.06    0.08    0.23    0.92 ^ clkbuf_leaf_47_CLK/X (sky130_fd_sc_hd__clkbuf_16)
                0.08    0.00    0.93 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    0.93   clock reconvergence pessimism
                         -0.03    0.89   library hold time
                                  0.89   data required time
---------------------------------------------------------------------------
                                  0.89   data required time
                                 -1.22   data arrival time
---------------------------------------------------------------------------
                                  0.33   slack (MET)


=== SUMMARY ===
worst slack max 40.13
worst slack min 0.33
tns max 0.00
wns max 0.00
Clock clk
   0.85 source latency core.CPU_Xreg_value_a5[17][8]$_DFF_P_/CLK ^
  -0.67 target latency core.OUT[8]$_DFF_P_/CLK ^
   0.00 CRPR
--------------
   0.18 setup skew

max slew

Pin                                   Limit    Slew   Slack
------------------------------------------------------------
core.OUT[8]$_DFF_P_/Q                  1.50    1.01    0.49 (MET)

max capacitance
```

```
Pin                                  Limit     Cap   Slack
-----------------------------------------------------------
_08069_/Y                             0.03    0.02    0.01 (MET)
```